

Instructions

1. You will need to have your video camera on so that I can see you while we discuss.
2. You will also need to share your screen for the duration of the exam.
3. You may access written material during the exam, but not anything on an electronic device.
4. This is a formal assessment, and is bound to the same rules on academic integrity as any other assessments for this class.

Questions

Below, you will find a series of short essay questions. For each short essay question, give your answer in the reserved space in the script.

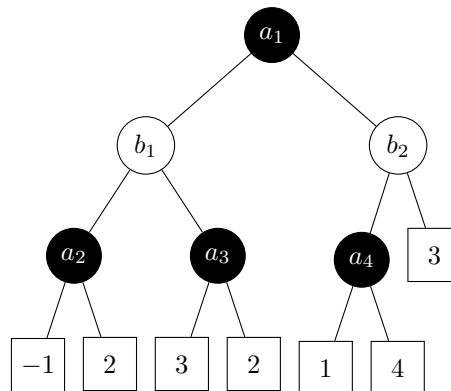
1. (25 points) Consider a search problem with a start node s and a single goal node t (states, costs, actions and state transitions are defined in the standard manner).

Let C^* be the cost of the optimal path from the start node s to the goal node t .

We run A^* graph search with the consistent heuristic h on this search problem.

- (a) (10 points) Consider a node n that was *not* explored during the search. We know that the cost to reach n from the start node s , referred to as $g(n)$, was $\frac{C^*}{2}$. Is it possible that $h(n) < \frac{C^*}{2}$? Explain your answer.
 - (b) (10 points) Now, consider a node n that was explored during the search. We know that the cost to reach n from the start node s , referred to as $g(n)$, was $\frac{C^*}{3}$. Is it possible that $h(n) > \frac{2C^*}{3}$? Explain your answer.
2. (25 points) Consider the minimax search tree shown below. Black nodes are controlled by the MAX player, and white nodes are controlled by the MIN player. Payments (terminal nodes) are squares; the number within denotes the amount that the MIN player pays to the MAX player. MAX wants to maximize the amount they receive, and MIN wants to minimize the amount they pay.

Consider the tree described in the diagram below and answer the following questions.



- (a) (5 points) Identify the subgame-perfect Nash equilibrium strategy of this game. Mark with an **X** every edge selected in an equilibrium strategy.
- (b) (10 points) Suppose that we run the α - β pruning algorithm on this tree. We check nodes from **right to left**. Mark with an **X** any branches that are pruned by the α - β pruning algorithm in the diagram below. In addition, write the final α and β values of each node next to the node (do not write α and β values for pruned branches).
- (c) (10 points) Suppose that we can modify the payoff of one of the nodes to whatever value we like. Which node's value should we modify, if our goal is to minimize the number of branches we prune?

3. (25 points) We have two players, 1 and 2. Each player can choose to play either L or R . Player 1 chooses first, followed by player 2. The payoffs to the players are as follows:

$$(L, L) \rightarrow (1, 1)$$

$$(L, R) \rightarrow (4, 0)$$

$$(R, L) \rightarrow (2, 1)$$

$$(R, R) \rightarrow (3, 2)$$

- (a) (10 points) Write this setting as an extensive form game. What is the subgame perfect Nash equilibrium?
- (b) (10 points) Suppose that player 2 cannot tell whether player 1 picked L or R . What is the Nash equilibrium in that case?
- (c) (5 points) Is it better for player to know whether player 1 picked L or R ?
4. (25 points) To showcase the relevance of our course material, let us consider the following recent event. In the 2026 World Cup group stage, we know the following facts.

C1: If it is not the case that team Iran beats team Egypt or team Croatia loses to team Ghana, or team DR Congo does not beat team Uzbekistan or team Algeria does not draw with team Austria, then team Iran does not advance.

C2: If the Iranian player is on offside and the VAR referee was used, then Team Iran does not beat team Egypt.

C3: The Iranian player was on offside.

C4: the VAR referee was used.

C5: Team Croatia did not lose to team Ghana

C6: Team Congo beat (did not lose or tie) team Uzbekistan

C7: Team Algeria tied with team Austria

Write the conditions above as sentences in first order logic, using only the following variables:

1. IA (team Iran advances)
2. IwE (team Iran beats team Egypt)
3. ClG (team Croatia loses to team Ghana)
4. CwU (team DR Congo beats team Uzbekistan)
5. AdA (team Algeria ties with team Austria)

Using a first-order logic proof technique studied in class, prove that team Iran did not advance.

END OF EXAM PAPER
